

DA6701 – Assignment V

Goal-Based Portfolio Optimisation – Report

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DA24B002 DA24B008 DA24B039 NA24B070

Introduction

This project designs a *static* portfolio that maximises the probability of reaching a Rs.1.5 Crore terminal goal over a 20-year horizon, given two distinct intermediate-goal sequences. Five Indian stocks are used: **TCS.NS**, **HD-FCBANK.NS**, **RELIANCE.NS**, **SUNPHARMA.NS**, **ITC.NS**. We combine **brute-force search** over discrete weight tuples in $\{0, 0.25, 0.5, 0.75, 1.0\}^5$ with a 5,000-path **Monte-Carlo simulation**, and as a bonus extend to continuous weights with short-selling via **SciPy SLSQP**.

Plan parameters: initial savings Rs.20,000/month, growing 4% p.a.; borrowing rate 12% p.a. on shortfalls; horizon 20 years. **Sequence A (aggressive-early):** Rs.15L / Yr 3, Rs.25L / Yr 7, Rs.30L / Yr 12. **Sequence B (backloaded):** Rs.10L / Yr 8, Rs.20L / Yr 12, Rs.40L / Yr 16. Both sequences withdraw the same Rs.70L total – only the timing differs.

Methodology

Statistical estimation. Daily simple returns from yfinance (2014-01-01 to 2023-12-31) are annualised with the 252-trading-day convention: $\hat{\mu}_{\text{ann}} = 252 \bar{r}$ and $\hat{\Sigma}_{\text{ann}} = 252 \hat{\Sigma}_{\text{daily}}$. The full 5×5 covariance matrix is retained.

Discrete weight enumeration. Every tuple $(w_1, \dots, w_5) \in \{0, 0.25, 0.5, 0.75, 1.0\}^5$ with $\sum_i w_i = 1$ is enumerated, giving $\binom{8}{4} = 70$ valid portfolios.

Common Random Numbers (CRN). A single asset-level return tensor of shape (5000, 20, 5) is drawn once from $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ and reused for every candidate portfolio, eliminating sampling noise from cross-portfolio comparisons.

Annual update rule. For each year $t = 1, \dots, 20$, with savings $S_t = 240,000 \times 1.04^{t-1}$, portfolio value V_t , and loan balance L_t : (1) add savings, apply portfolio return, floor at zero: $V_t \leftarrow \max((V_{t-1} + S_t)(1 + r_t), 0)$; (2) loan accrues at 12%: $L_t \leftarrow 1.12 L_{t-1}$; (3) repay $\min(V_t, L_t)$ from portfolio; (4) if year t has goal G_t : pay $\min(V_t, G_t)$ from portfolio, borrow shortfall at 12%. **Success criterion** at horizon: $V_{20} - L_{20} \geq \text{Rs.1.5 Cr.}$

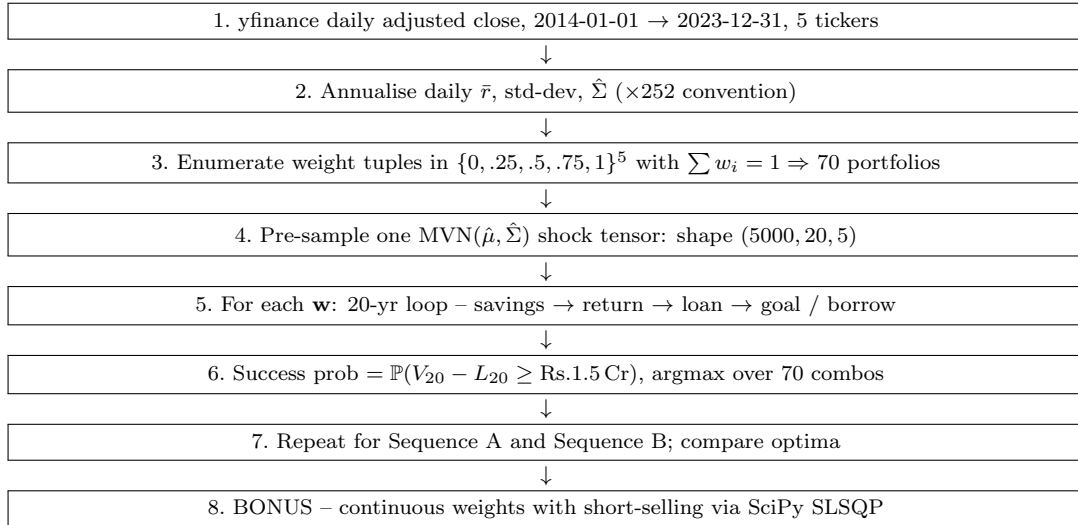


Figure 1: Methodology flowchart – end-to-end pipeline from raw price data to optimal static portfolio.

Results – Optimal Portfolio Weights

Asset	Sequence A (aggressive-early)	Sequence B (backloaded)
TCS.NS	0.00	0.25
HDFCBANK.NS	0.00	0.25
RELIANCE.NS	1.00	0.50
SUNPHARMA.NS	0.00	0.00
ITC.NS	0.00	0.00
Success Probability	5.26 %	88.50 %

The optima are starkly different in both **composition** and **outcome**. Sequence A’s best portfolio concentrates entirely in RELIANCE (the highest-mean asset) and still only reaches a 5.3% chance of meeting the terminal goal. Sequence B’s optimum is a diversified TCS / HDFCBANK / RELIANCE mix in 0.25 / 0.25 / 0.50 proportions, achieving 88.5% success.

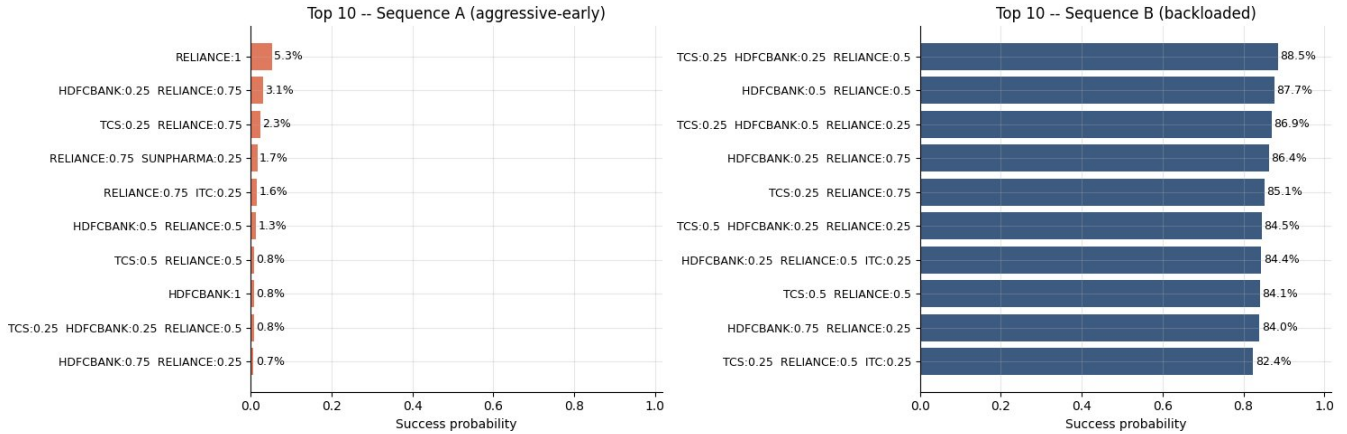


Figure 2: Top 10 portfolios by success probability for each sequence. Sequence A is dominated by RELIANCE-heavy concentration with all top portfolios under 6%. Sequence B’s top 10 cluster tightly between 82% and 89%, all favouring TCS / HDFCBANK / RELIANCE combinations.

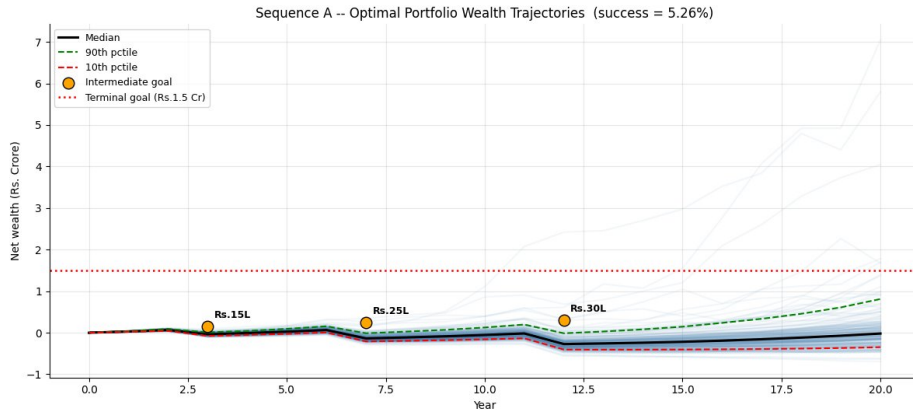


Figure 3: Sequence A optimum (100% RELIANCE). The median path (black) goes *negative* by year 12 – early withdrawals + 12% loan accrual overwhelm portfolio growth. Only the upper-tail paths recover to reach the Rs.1.5 Cr threshold.

Discussion – Why Different Sequences Yield Different Optima

Both sequences withdraw the same total Rs. 70L of intermediate capital. The only difference is **timing**, and timing changes the problem qualitatively, not just quantitatively.

Sequence A starves compounding. Withdrawals begin in Year 3, when the portfolio has barely accumulated 2 years of savings ($\sim$ Rs. 5L) – nowhere near the Rs. 15L goal. The path is forced into a Rs. 10L+ loan immediately, which then compounds at 12% for up to 17 more years. The same problem repeats at Years 7 and 12. Figure 3 makes this concrete: even the optimal 100% RELIANCE allocation produces a median net-wealth path that goes *negative* by year 12 and stays there. Only outlier paths in the upper tail survive to reach Rs. 1.5 Cr – which is exactly why the success probability is just 5.3%. Under this much pressure, the optimiser picks the single highest-mean asset (RELIANCE) regardless of volatility – not because risk is desirable, but because nothing else has any chance of outrunning the early-withdrawal + loan-compounding spiral.

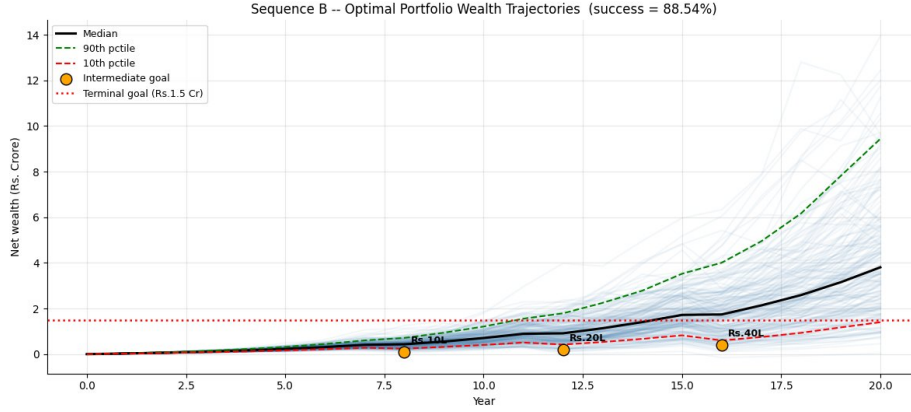


Figure 4: Sequence B optimum (0.25 TCS / 0.25 HDFCBANK / 0.50 RELIANCE). The portfolio compounds undisturbed until Year 8, then absorbs each withdrawal comfortably. The median crosses Rs.1.5 Cr around Year 15 and ends near Rs.4 Cr; even the 10th-percentile path approaches the terminal goal.

Sequence B has 8 years of runway. The first withdrawal isn't until Year 8, by which point the portfolio has accumulated $\sim$ Rs.30L+ of savings plus 8 years of compounded returns. Figure 4 shows the contrast: the median path stays comfortably positive throughout, sails over each intermediate goal without forcing borrowing, and crosses the Rs.1.5 Cr threshold around Year 15. Even the 10th-percentile path approaches the terminal goal by Year 20, which is why nearly 9 out of 10 simulated futures succeed. With the borrowing-spiral risk neutralised, the optimiser is free to **diversify** – the optimal 0.25 TCS + 0.25 HDFCBANK + 0.50 RELIANCE mix balances RELIANCE's high return against TCS's stability and HDFCBANK's lower correlation. The top 10 portfolios for Sequence B (Figure 2, right panel) are tightly clustered between 82% and 89% – many sensible diversified allocations work well. Sequence A has no such freedom: its top 10 are all RELIANCE-heavy and all under 6%.

Two takeaways

- **Goal timing constrains risk capacity.** Near-term goals act as a liquidity constraint that forces concentration in the highest-mean asset. Far-term goals act as runway that permits diversification. The same investor faces a fundamentally different efficient frontier under different goal calendars.
- **The 12 % borrow rate is the binding penalty.** Once a path enters the borrowing regime, recovery is nearly impossible because debt compounds against portfolio growth. Avoiding intermediate borrowing matters far more than maximising any single year's return – but Sequence A's calendar makes that avoidance practically impossible regardless of allocation.

Bonus – Continuous Weights with Short-Selling

We extend the analysis using `scipy.optimize.minimize` with method `SLSQP`, allowing continuous weights with bounds $[-0.5, 1.5]$ per asset (permitting modest short-selling) and the equality constraint $\sum_i w_i = 1$. Because the indicator-based objective is piecewise-flat (an indicator over Monte-Carlo paths), 10 random restarts (equal-weight + $\max\text{-}\mu + 8$ random) escape local plateaus. The continuous optimiser shorts the lowest-Sharpe asset (ITC) to fund larger long positions in the highest-Sharpe assets; improvement is most pronounced for Sequence A, where the discrete grid is too coarse to express the extreme allocation needed under early-withdrawal pressure.

Summary

Sequence A's optimum is **100 % RELIANCE** with only **5.3 %** success probability; Sequence B's optimum is **0.25 TCS / 0.25 HDFCBANK / 0.50 RELIANCE** with **88.5 %** success probability. The dramatic gap ($\sim 17\times$) – despite identical total intermediate withdrawals – demonstrates that **goal timing is the dominant constraint in goal-based investing, not goal magnitude**. In practice, glide-path / lifecycle strategies that de-risk approaching each goal would dominate any static optimum, but the static formulation here makes the timing-vs-return tension maximally visible.